

Parallel And Gpu Computing

Lab Experiment 1

Sequential Matrix Multiplication

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Aim

To write and run a C program that multiplies two 4000 x 4000 matrices sequentially (on a single CPU), using WSL2 Ubuntu, and to note down the execution time.

Software Used

Windows PowerShell, WSL2 (Ubuntu), GCC compiler.

What I Did

1. First I opened Windows PowerShell and checked whether WSL was installed properly on my system:

```
wsl --status  
wsl -l -v
```

```
PS C:\WINDOWS\system32> wsl --status  
Default Distribution: Ubuntu  
Default Version: 2  
WSL1 is not supported with your current machine configuration.  
Please enable the "Windows Subsystem for Linux" optional component to use WSL1.  
PS C:\WINDOWS\system32> wsl -l -v  
NAME      STATE      VERSION  
* Ubuntu   Stopped    2
```

2. Then I started WSL and checked my current directory:

```
wsl  
pwd
```

```
PS C:\WINDOWS\system32> wsl --install -d ubuntu  
A distribution with the supplied name already exists. Use --name to chose a different name.  
Error code: Wsl/InstallDistro/ERROR_ALREADY_EXISTS  
PS C:\WINDOWS\system32> wsl -l -v  
NAME      STATE      VERSION  
* Ubuntu   Stopped    2  
PS C:\WINDOWS\system32> wsl  
dheeraj@DESKTOP-RIAR5TT:/mnt/c/WINDOWS/system32$ cd ~  
dheeraj@DESKTOP-RIAR5TT:~$ pwd  
/home/dheeraj
```

3. Inside Ubuntu, I updated the package list first:

```
cd ~  
sudo apt update
```

4. I installed build-essential (for GCC) and checked the compiler version:

```
sudo apt install build-essential -y  
gcc --version
```

```

dheeraj@DESKTOP-RIAR5TT:~$ ls
matrix Sequential matrix Sequential.c
dheeraj@DESKTOP-RIAR5TT:~$ sudo apt update
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
Hit:1 http://archive.ubuntu.com/ubuntu InRelease
Get:2 http://security.ubuntu.com/ubuntu InRelease [137 kB]
Get:3 http://archive.ubuntu.com/ubuntu InRelease [137 kB]
Get:4 http://archive.ubuntu.com/ubuntu InRelease [137 kB]
Get:5 http://security.ubuntu.com/ubuntu InRelease [471 kB]
Get:6 http://archive.ubuntu.com/ubuntu InRelease [603 kB]
Get:7 http://security.ubuntu.com/ubuntu InRelease [110 kB]
Get:8 http://security.ubuntu.com/ubuntu InRelease [46.6 kB]
Get:9 http://archive.ubuntu.com/ubuntu InRelease [143 kB]
Get:10 http://security.ubuntu.com/ubuntu InRelease [171 kB]
Get:11 http://archive.ubuntu.com/ubuntu InRelease [98.3 kB]
Get:12 http://archive.ubuntu.com/ubuntu InRelease [285 kB]
Get:13 http://security.ubuntu.com/ubuntu InRelease [43.4 kB]
Get:14 http://archive.ubuntu.com/ubuntu InRelease [91.0 kB]
Get:15 http://security.ubuntu.com/ubuntu InRelease [399 kB]
Get:16 http://archive.ubuntu.com/ubuntu InRelease [188 kB]
Get:17 http://archive.ubuntu.com/ubuntu InRelease [409 kB]
Get:18 http://security.ubuntu.com/ubuntu InRelease [77.7 kB]
Get:19 http://archive.ubuntu.com/ubuntu InRelease [79.5 kB]
Get:20 http://security.ubuntu.com/ubuntu InRelease [10.8 kB]
Get:21 http://archive.ubuntu.com/ubuntu InRelease [11.6 kB]
Get:22 http://security.ubuntu.com/ubuntu InRelease [2844 B]
Get:23 http://archive.ubuntu.com/ubuntu InRelease [3036 B]
Get:24 http://archive.ubuntu.com/ubuntu InRelease [1056 B]
Fetched 3657 kB in 7s (544 kB/s)
155 packages can be upgraded. Run 'apt list --upgradable' to see them.

```

```

Select dheeraj@DESKTOP-RIAR5TT: ~/pgcu113/seq
155 packages can be upgraded. Run 'apt list --upgradable' to see them.
dheeraj@DESKTOP-RIAR5TT:~$ sudo apt install build-essential -y
build-essential is already the newest version (12.12ubuntu2.26.04.2).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 155
dheeraj@DESKTOP-RIAR5TT:~$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

```

5. I made a separate folder for this program so it doesn't mix with anything else:

```

mkdir -p ~/pgcu113/seq
cd ~/pgcu113/seq

```

6. I created the C file using nano and typed the program:

```

dheeraj@DESKTOP-RIAR5TT:~/pgcu113/matrix_seq$ mkdir -p ~/pgcu113/seq
dheeraj@DESKTOP-RIAR5TT:~/pgcu113/matrix_seq$ cd ~/pgcu113/seq
dheeraj@DESKTOP-RIAR5TT:~/pgcu113/seq$ pwd
/home/dheeraj/pgcu113/seq
dheeraj@DESKTOP-RIAR5TT:~/pgcu113/seq$ ls
dheeraj@DESKTOP-RIAR5TT:~/pgcu113/seq$ nano matrix Sequential113.c
dheeraj@DESKTOP-RIAR5TT:~/pgcu113/seq$ nano matrix Sequential113.c

```

nano matrix Sequential113.c

C code for Sequential Matrix Multiplication

```
#include <stdio.h>
#include <stdlib.h>
#include <time.h>

#define N 4000

int main()
{
    int i, j, k;
    double *A, *B, *C;
    clock_t start, end;

    A = (double *)malloc(N * N * sizeof(double));
    B = (double *)malloc(N * N * sizeof(double));
    C = (double *)malloc(N * N * sizeof(double));

    if (A == NULL || B == NULL || C == NULL)
    {
        printf("Memory allocation failed\n");
        return 1;
    }

    printf("Initializing %d x %d matrices...\n", N, N);
    for (i = 0; i < N; i++)
    {
        for (j = 0; j < N; j++)
        {
            A[i * N + j] = 1.0;
            B[i * N + j] = 1.0;
            C[i * N + j] = 0.0;
        }
    }

    start = clock();
    for (i = 0; i < N; i++)
    {
        for (j = 0; j < N; j++)
        {
            for (k = 0; k < N; k++)
            {
                C[i * N + j] += A[i * N + k] * B[k * N + j];
            }
        }
    }
    end = clock();

    printf("\nSequential Matrix Multiplication Completed\n");
    printf("Matrix Size = %d x %d\n", N, N);
    printf("Execution Time = %f seconds\n",
           (double)(end - start) / CLOCKS_PER_SEC);
    printf("Verification C[0][0] = %.2f\n", C[0]);

    free(A);
    free(B);
    free(C);
    return 0;
}
```

```
dheeraj@DESKTOP-RIAR5TT: ~/pgcu113/seq
GNU nano 8.7.1 matrix_sequential113.c
#include <stdio.h>
#include <stdlib.h>
#include <time.h>

#define N 4000

int main()
{
    int i, j, k;
    double *A, *B, *C;
    clock_t start, end;

    A = (double *)malloc(N * N * sizeof(double));
    B = (double *)malloc(N * N * sizeof(double));
    C = (double *)malloc(N * N * sizeof(double));

    if (A == NULL || B == NULL || C == NULL)
    {
        printf("Memory allocation failed\n");
        return 1;
    }

    printf("Initializing %d x %d matrices...\n", N, N);

    for (i = 0; i < N; i++)
    {
        for (j = 0; j < N; j++)
        {
            A[i * N + j] = 1.0;
            B[i * N + j] = 1.0;
            C[i * N + j] = 0.0;
        }
    }

    start = clock();

    for (i = 0; i < N; i++)
    {
        for (j = 0; j < N; j++)
        {
            for (k = 0; k < N; k++)
            {
                C[i * N + j] +=
                    A[i * N + k] *
                    B[k * N + j];
            }
        }
    }

    end = clock();

    printf("\nSequential Matrix Multiplication Completed\n");
    printf("Matrix Size = %d x %d\n", N, N);
    printf("Execution Time = %f seconds\n",
        (double)(end - start) / CLOCKS_PER_SEC);
    printf("Verification C[0][0] = %.2f\n", C[0]);

    free(A);
    free(B);
    free(C);

    return 0;
}
```

GNU nano 8.7.1 matrix_sequential113.c

Help Write Out Where Is Cut Execute Location M-U Undo
Exit Read File Replace Paste Justify Go To Line M-E Redo

7. After saving the file, I compiled it using GCC with the -O2 optimization flag:

```
gcc -O2 matrix_sequential113.c -o matrix_sequential113
```

8. I checked that the executable file got created:

```
ls -l
```

9. Finally, I ran the program:

```
./matrix_sequential113
```

```
dheeraj@DESKTOP-RIAR5TT:~/pgcu113/seq$ ls
dheeraj@DESKTOP-RIAR5TT:~/pgcu113/seq$ nano matrix_sequential113.c
dheeraj@DESKTOP-RIAR5TT:~/pgcu113/seq$ nano matrix_sequential113.c
dheeraj@DESKTOP-RIAR5TT:~/pgcu113/seq$ gcc -O2 matrix_sequential113.c -o matrix_sequential113
dheeraj@DESKTOP-RIAR5TT:~/pgcu113/seq$ ls -l
total 20
-rwxr-xr-x 1 dheeraj dheeraj 16184 Sep 11 04:57 matrix_sequential113
-rw-r--r-- 1 dheeraj dheeraj 1283 Sep 11 04:55 matrix_sequential113.c
dheeraj@DESKTOP-RIAR5TT:~/pgcu113/seq$ ./matrix_sequential113
Initializing 4000 x 4000 matrices...

Sequential Matrix Multiplication Completed
Matrix Size = 4000 x 4000
Execution Time = 312.342084 seconds
Verification C[0][0] = 4000.00
```

Output

Matrix Size = 4000 x 4000

Execution Time = 312.342084 seconds

Verification $C[0][0] = 4000.00$

Result

I was able to successfully compile and run the sequential matrix multiplication program in WSL2 Ubuntu. The output $C[0][0] = 4000.00$ matched the expected value, so the program worked correctly. It took 312.342084 seconds to multiply the two 4000 x 4000 matrices sequentially.